LESSON 1.5

Math is Finding Patterns

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find patterns and pattern rules, use them to make generalizations, and check my solutions with a table of values.

WHY THIS MATTERS Akela's change jar is full of pennies, nickels, dimes, and quarters. The value of the coins is $53.50. How many of each coin could be in the jar? There is more than one answer — and patterns are the way in.

REMEMBER

Math is Finding Patterns

Math Practices: Patterns & Generalizations The value of each coin is a pattern. Noticing patterns — and the calculations that repeat — lets us solve problems we could never solve by counting one coin at a time.

1. Organize observations into a table of values.
2. Identify the constant change or rule connecting inputs to outputs.
3. Write a general rule or expression to predict future terms

Words: **pattern *(patrón)*** **pattern rule *(regla del patrón)*** **generalization *(generalización)*** **table of values *(tabla de valores)*** **reasonableness *(sensatez del resultado)***

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each pattern: add the same amount, or multiply by the same amount?

Groups: **Add the same amount each time** **Multiply by the same amount each time**

_____ 4, 6, 8, 10, ...

_____ 18, 20, 22, 24, ...

_____ 4.5, 7.0, 9.5, 12.0, ...

_____ 2, 4, 8, 16, ...

_____ 1, 3, 9, 27, ...

_____ 5, 50, 500, 5000, ...

HINT Check the jump between neighbors: is it the same DIFFERENCE, or the same FACTOR?

2 **Complete the table. Show how you found each value.**

Player A starts with 18 points and makes two-pointers. Complete the table.

Two-pointers made	Rule: 18 + 2 × shots	Total points
1		
5		
10		

HINT Multiply first, then add the starting 18 points.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

Which box holds more — one that holds 20 cubic inches, or one that holds 30 cubic inches?

(A) The 20 cubic inch box

(B) The 30 cubic inch box

(C) They hold the same

(D) You cannot tell

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A cube is 2 cm long on every edge. What is its volume?

(A) 4 cubic cm
(B) 6 cubic cm
(C) 8 cubic cm
(D) 12 cubic cm

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

5 **Circle the letter of the best answer. Show how you know.**

A box is 2 cm long, 3 cm wide, and 2 cm tall. What is its volume?

(A) 7 cubic cm
(B) 10 cubic cm
(C) 12 cubic cm
(D) 24 cubic cm

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

EXPLAIN YOUR THINKING

Take a card you called 'A one-time fact.' What would have to keep happening for it to become a generalization instead — and how would you write that as a pattern rule?

SENTENCE STARTER ___ is only a one-time fact because ___; it would be a pattern if ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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LEARNING TARGET I can find patterns and pattern rules, use them to make generalizations, and check my solutions with a table of values.

WHY THIS MATTERS Akela's change jar is full of pennies, nickels, dimes, and quarters. The value of the coins is $53.50. How many of each coin could be in the jar? There is more than one answer — and patterns are the way in.

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

Which statement best captures what this lesson means by 'Math is Finding Patterns'?

(A) Patterns are decorations that make problems look nicer

(B) A pattern rule only works if you also count every item one at a time

(C) We look for patterns and relationships, use them to solve problems and make generalizations, and check our solutions for reasonableness

(D) Once you spot a pattern you never need to check your answer

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Two basketball players make 10 shots in a row after scoring some points. Player A had scored 18 points before making 10 two-pointers in a row. Player B had 6 points before scoring 10 three-pointers in a row. Player A had scored 18 points before making 10 two-pointers in a row. Using the pattern rule of adding 2 points per shot, what is Player A's final score?

(A) 20
(B) 28
(C) 36
(D) 38

PER 1

3 **Circle the letter of the best answer. Show how you know.**

Two basketball players make 10 shots in a row after scoring some points. Player A had scored 18 points before making 10 two-pointers in a row. Player B had 6 points before scoring 10 three-pointers in a row. Player B had 6 points before scoring 10 three-pointers in a row. Using the pattern rule of adding 3 points per shot, what is Player B's final score?

(A) 16
(B) 30
(C) 36
(D) 38

PER 1

4 **Circle the letter of the best answer. Show how you know.**

Two basketball players make 10 shots in a row after scoring some points. Player A had scored 18 points before making 10 two-pointers in a row. Player B had 6 points before scoring 10 three-pointers in a row. After using the pattern rules to find both players' final scores, how can the players check that their totals are reasonable?

(A) Multiply the two final scores together

(B) Round both totals to the nearest ten

(C) List the score after each shot in a table of values and confirm the sequence lands on the final totals

(D) Compare the players' shooting percentages

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Take a card you called 'A one-time fact.' What would have to keep happening for it to become a generalization instead — and how would you write that as a pattern rule?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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